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ASSIGNMENT No: 14

TITLE: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives : To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory: Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Exercise

1. Create an ATM program handling invalid withdrawal amount using try-catch

```
 import java.util.Scanner;


public class ATM {
    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);


        double balance = 10000;


        try {
```

```
            System.out.print("Enter withdrawal amount: ");
            double amount = sc.nextDouble();

            if (amount <= 0)
                throw new Exception("Invalid withdrawal amount!");

            if (amount > balance)
                throw new Exception("Insufficient balance!");

            balance -= amount;

            System.out.println("Withdrawal Successful.");
            System.out.println("Remaining Balance: " + balance);

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

```
import java.util.Scanner;

public class OnlineShopping {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
        try {
            System.out.print("Enter product name: ");
            String product = sc.nextLine();

            System.out.print("Enter quantity: ");
            int quantity = sc.nextInt();

            if (quantity <= 0)
                throw new Exception("Invalid quantity! Quantity must be greater than
zero.");

            System.out.println("\nOrder Placed Successfully!");
            System.out.println("Product: " + product);
            System.out.println("Quantity: " + quantity);

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```